

Application-Level Logging Tools

1. Overview

In production systems, logs are generated by:

- Applications
- APIs
- Databases
- Containers
- OS / Infra
- Security devices

As traffic grows, logs become huge.

We need centralized tools to:

- Collect logs
- Store logs
- Search logs
- Visualize trends
- Alert on errors

This is where **logging platforms** are used.

2. Why Centralized Logging is Important

Without centralized logging:

None

Server-1 logs in file A

Server-2 logs in file B

Container logs deleted

No correlation

Very hard to debug.

With centralized logging:

None

All logs → One platform

Search by request_id

Find errors in seconds

3. Popular Logging Stack: ELK

ELK stands for:

- Elasticsearch
- Logstash
- Kibana

Later expanded to **Elastic Stack** with Beats.

4. ELK Components

A. Logstash

Used for:

- Collecting logs
- Parsing logs
- Transforming data
- Routing logs

Example:

None

App Logs → Logstash → Elasticsearch

B. Elasticsearch

Distributed search engine.

Used for:

- Indexing logs
- Full-text search
- Fast filtering
- Aggregations

Example:

Search:

None

`status=500 AND service=checkout`

C. Kibana

Frontend UI for:

- Dashboards
- Charts
- Searching logs
- Alerting (some setups)

D. Beats

Lightweight agents installed on hosts.

Examples:

- Filebeat → ship log files
- Metricbeat → system metrics
- Packetbeat → network data

5. ELK Data Flow

None

Application



Filebeat



Logstash



Elasticsearch



Kibana Dashboard

6. Splunk

Enterprise commercial platform.

Features:

- Powerful search
- Security analytics
- Dashboards
- Alerting
- Enterprise support

Used in large enterprises.

Tradeoff:

- Expensive at scale.

7. Monitoring vs Logging Tools

Some tools focus on logs, some on metrics.

Tool	Best For
ELK	Logging
Splunk	Enterprise logging
Prometheus	Metrics
Grafana	Dashboards

Datadog	Full observability
New Relic	APM + monitoring

8. Prometheus + Grafana

Very common in modern HLD.

Prometheus

Collects metrics by scraping endpoints.

None

`/metrics`

`cpu_usage 65`

`http_requests_total 12000`

Uses **PromQL** query language.

Grafana

Used for:

- Beautiful dashboards
- Graphs
- Multi-source data visualization

9. Alertmanager

Used with Prometheus.

Example:

None

If P99 > 1s for 5 mins

→ PagerDuty / Email / Slack

10. Time Series Database (TSDB)

Metrics are time-based data.

Examples:

None

10:00 CPU 45%

10:01 CPU 50%

10:02 CPU 63%

Stored efficiently in TSDB.

Popular TSDBs:

- Prometheus TSDB
- InfluxDB
- Graphite
- OpenTSDB

11. Why TSDB Needed

Optimized for:

- Continuous writes
- Time-window queries
- Aggregation
- Retention policies

12. Downsampling

Old detailed data may be expensive.

Instead of keeping per-second metrics forever:

Keep:

- 1 sec granularity for 7 days
- 1 min averages for 30 days
- 1 hour averages for 1 year

This reduces cost.

13. Cold Storage

Very old logs can move to cheaper storage:

- Object storage
- Tape archives
- Glacier-like storage

Useful for compliance.

14. Tool Selection Criteria

Interview gold answer.

Choose tools based on:

Functional

- Logging support
- Metrics support
- Alerting
- Tracing integration

Operational

- Scalability
- Reliability
- Resource usage

Human Factors

- Easy UI
- Learning curve
- Engineer availability

Cost

- License cost
- Storage cost
- Compute cost

15. Common Problems in Logging Systems

- Huge storage bills
- Noisy logs
- Sensitive data leakage
- Slow search queries
- Missing retention policy
- Logs without request IDs

16. Security Best Practices

Never log:

- Passwords
- Tokens
- Card numbers
- PII unless masked

Use RBAC for log access.

17. Interview Script

Say:

“I’d use centralized logging with ELK or Splunk, metrics with Prometheus, dashboards via Grafana, and alerts through Alertmanager/PagerDuty. I’d enforce retention policies and structured logs with request IDs.”

Strong answer.

18. Final Summary

- Centralized logging is essential at scale
- ELK = collect, search, visualize logs
- Prometheus = metrics collection
- Grafana = dashboards
- TSDB stores time-series efficiently
- Downsampling reduces storage cost
- Alerts convert signals into action

19. Memorable Line

Logs tell stories. Metrics tell trends. Traces tell journeys.